Project Workflow

The AI Flood Prediction System follows a structured machine learning workflow consisting of five major epics. Each epic represents a specific phase of the project, starting from data collection and ending with the deployment of the flood prediction web application.

Epic 1: Data Collection

Story 1: Collect the flood prediction dataset from a reliable source and load it into the Python environment (Jupyter Notebook/PyCharm) for analysis and model development.

Epic 2: Visualizing and Analysing the Data

Story 1: Import the required Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, and XGBoost.

Story 2: Load the dataset and explore its structure, features, and target variable.

Story 3: Perform Univariate Analysis to understand the distribution of individual features.

Story 4: Perform Multivariate Analysis to identify relationships among different features.

Story 5: Conduct descriptive statistical analysis and visualize the data using graphs and charts.

Epic 3: Data Pre-processing

Story 1: Identify and handle missing values.

Story 2: Detect and remove outliers.

Story 3: Encode categorical features into numerical values.

Story 4: Split the dataset into training and testing datasets.

Story 5: Apply feature scaling to improve model performance.

Epic 4: Model Building

Story 1: Train the Decision Tree model.

Story 2: Train the Random Forest model.

Story 3: Train the K-Nearest Neighbours (KNN) model.

Story 4: Train the XGBoost model.

Story 5: Compare all models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Story 6: Select the best-performing model (XGBoost), achieving **96.55% accuracy**, and save it as a **Pickle (.pkl)** file for deployment.

Epic 5: Application Building

Story 1: Design responsive HTML pages using HTML, CSS, and Bootstrap.

Story 2: Develop the Flask web application and integrate the trained machine learning model.

Story 3: Test the application using different weather inputs and display the flood prediction results through a user-friendly interface.

Workflow Diagram:

Dataset Collection

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Data Visualization

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Data Pre-processing

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Model Training

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Model Evaluation

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Best Model (XGBoost)

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Save Model (.pkl)

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Flask Web Application

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Flood Prediction Result